

2024/06/07 電力工程導論 姓名_____學號_____

1. (20) 如圖假設三相導體(a,b,c)有相同距離D且相同導體半徑r, 假設電荷平衡 $q_a + q_b + q_c = 0$ (電容 $c_a = c_b = c_c = c$, 電壓 $v_a + v_b + v_c = 0$), 假設電流平衡

$i_a + i_b + i_c = 0$. (a) 求電容值 $c = ?$ (b) 求a相磁通鏈 $\lambda_a = ?$ (每相導體數1條或多條差異?)

$$V_a + V_b + V_c = 0, \quad i_a + i_b + i_c = 0$$

$$\lambda = L \times i \quad (\text{电感})$$

$$\Rightarrow V_a i_a + V_b i_b + V_c i_c \neq 0$$

$$C = \frac{q}{V} = \frac{2\pi\epsilon}{\ln(D/r)} \quad r: \text{conductor diameter}$$

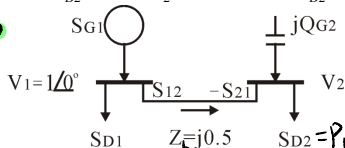
2. (20) 三相60-Hz 138-kV (線對線) 傳輸線長225mile (英哩). 傳輸線參數電阻 $r = 0.169 \Omega/\text{mile}$, 電感 $l = 2.093 \text{ mH}/\text{mile}$, 電容 $c = 0.01427 \mu\text{F}/\text{mile}$, 並聯電導 $g = 0$.

三相傳輸線傳送 40 Mwat 132 kV 於落後功因值95%. (a) 求發送端的電壓和電流.

(b) 求傳輸線效率. $\text{Chp. 4 p. 5} \quad \cos\theta = 0.95 \quad \theta \text{ 功因角} \quad \Rightarrow \vec{S}_1 \cdot \vec{V}_1 \rightarrow I_1$

3. (20) 利用電力傳送接收圓的觀念, 已知 $V_1 = 1 \angle 0^\circ$, $jQ_{G2} = j1.0$, $Z_L = j0.5$, S_{D2}

$= P_{D2} + j1.0$. 探討實功 P_{D2} 的操作可能性於 $P_{D2} \geq 0$. (a) 當實功 $P_{D2} > 1 \Rightarrow V_2 = ?$ (b) 當實功 $P_{D2} = 1 \Rightarrow V_2 = ?$ (c) 當實功 $P_{D2} = 0.5 \Rightarrow V_2 = ?$

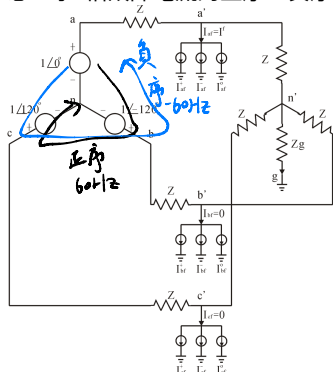


$$P_{12} = -P_{21} = \frac{|V_1||V_2|}{X} \sin\theta_{12} \quad X = \omega l$$

$$Q_{12} = \frac{|V_1|^2}{X} - \frac{|V_1||V_2|}{X} \cos\theta_{12}$$

$$Q_{21} = \frac{|V_2|^2}{X} - \frac{|V_1||V_2|}{X} \cos\theta_{12}$$

4. (20) 故障電流: $I^f = [I_{af} \ I_{bf} \ I_{cf}] = [I^f \ 0 \ 0]$ 發生於a相. 利用正序、負序和零序的觀念, 求a相故障電流的正序、負序和零序成分.



5. (20) $v_a = 180 \cos \omega t$, $v_b = 180 \cos (\omega t - 120^\circ)$, $v_c = 180 \cos (\omega t + 120^\circ)$ (a) 求abc參考座標轉換至靜止框座標的值 (b) 求靜止框座標轉換至同步框座標的值

$$\omega \begin{bmatrix} v_{gs} \\ v_{ds} \end{bmatrix} = \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ 0 & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix}$$

$$(b) \begin{bmatrix} V_{ge} \\ V_{be} \end{bmatrix} = \begin{bmatrix} \cos \omega t & -\sin \omega t \\ \sin \omega t & \cos \omega t \end{bmatrix} \begin{bmatrix} V_{gs} \\ V_{bs} \end{bmatrix}$$

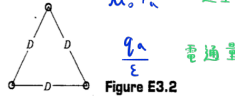
6.(20) 新的發電設備要與交流市電併網時的注意事項為何(列舉四個重點)

同電壓大小, 同頻率, 同相序, 同角度 (併網後可微調)
(轉速)

1. (b)

Example 3.2

Calculate the inductance per meter of each phase of a three-phase transmission line (Figure E3.2). Assume that



1. Conductors are equally spaced, D , and have equal radii r .
2. $i_a + i_b + i_c = 0$.

Solution Using (3.21) for phase a , we have

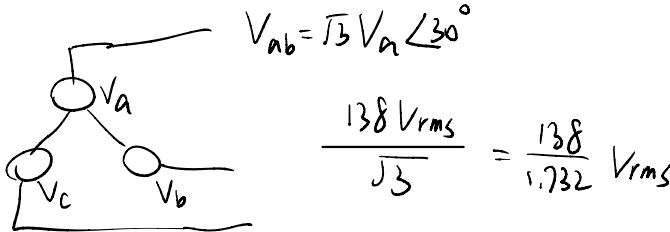
$$\begin{aligned} \lambda_a &= \frac{\mu_0}{2\pi} \left(i_a \ln \frac{1}{r'} + i_b \ln \frac{1}{D} + i_c \ln \frac{1}{D} \right) \\ &= \frac{\mu_0}{2\pi} \left(i_a \ln \frac{1}{r'} - i_b \ln \frac{1}{D} \right) \\ &= \frac{\mu_0}{2\pi} i_a \ln \frac{D}{r'} \end{aligned}$$

Thus,

$$L_a = \frac{\lambda_a}{i_a} = \frac{\mu_0}{2\pi} \ln \frac{D}{r'} = 2 \times 10^{-7} \ln \frac{D}{r'} \quad (3.23)$$

In the derivation use has been made of the equilateral spacing and the fact that $i_c = -(i_a + i_b)$. Because of symmetry we get the same results for L_b and L_c . Thus, we get the self-inductances $L_a = L_b = L_c = (\mu_0/2\pi) \ln(D/r')$ henrys per meter. We also note that L_a depends on i_a alone (i.e., there are no mutual inductance terms in this particular description). The same is true for b and c phases.

2.



$$Z = r + j\omega l$$

series imped. per meter

$$Y = g + j\omega c$$

shunt admittance per meter

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